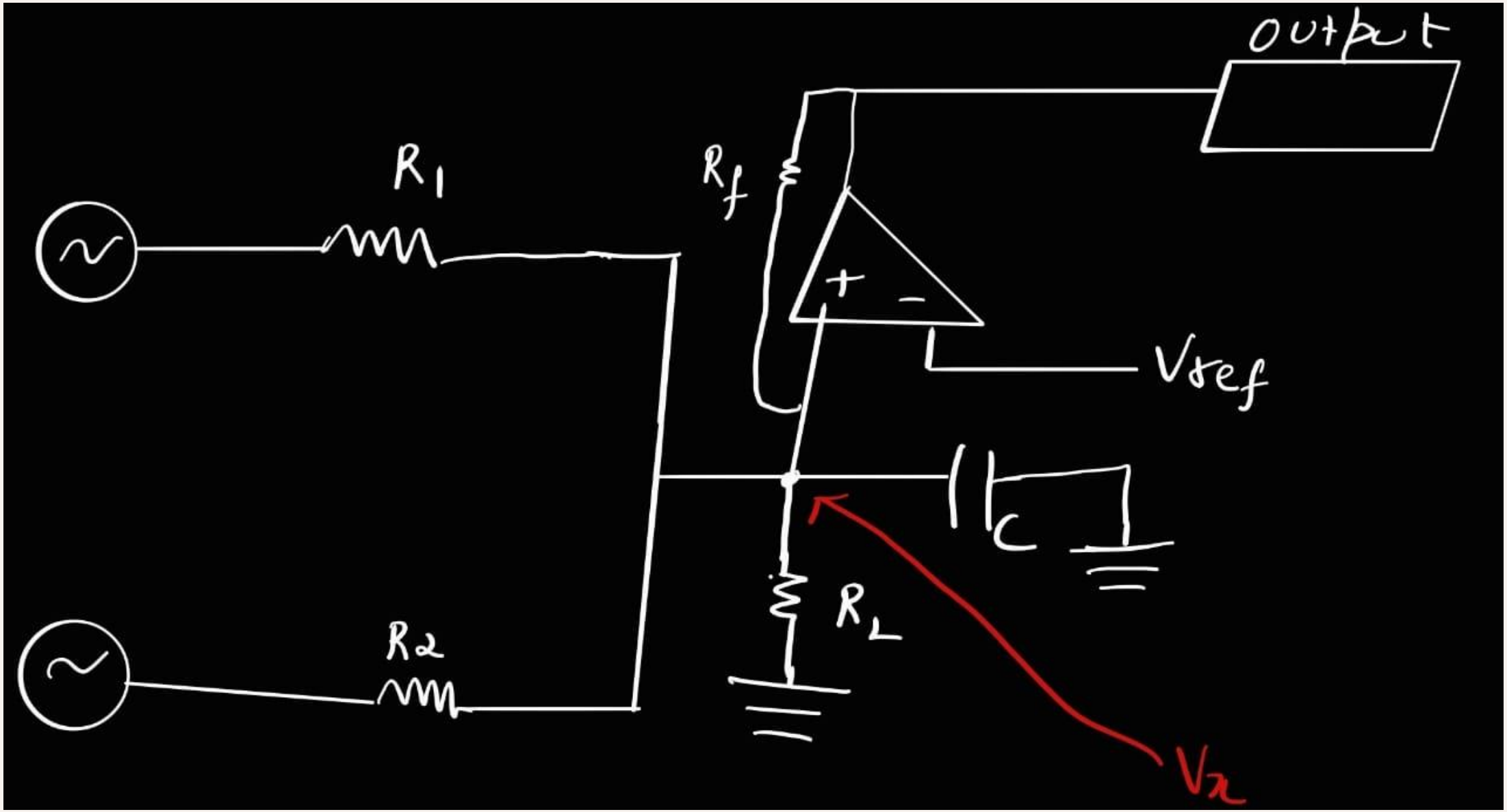
PHASE-2

SIMULATION

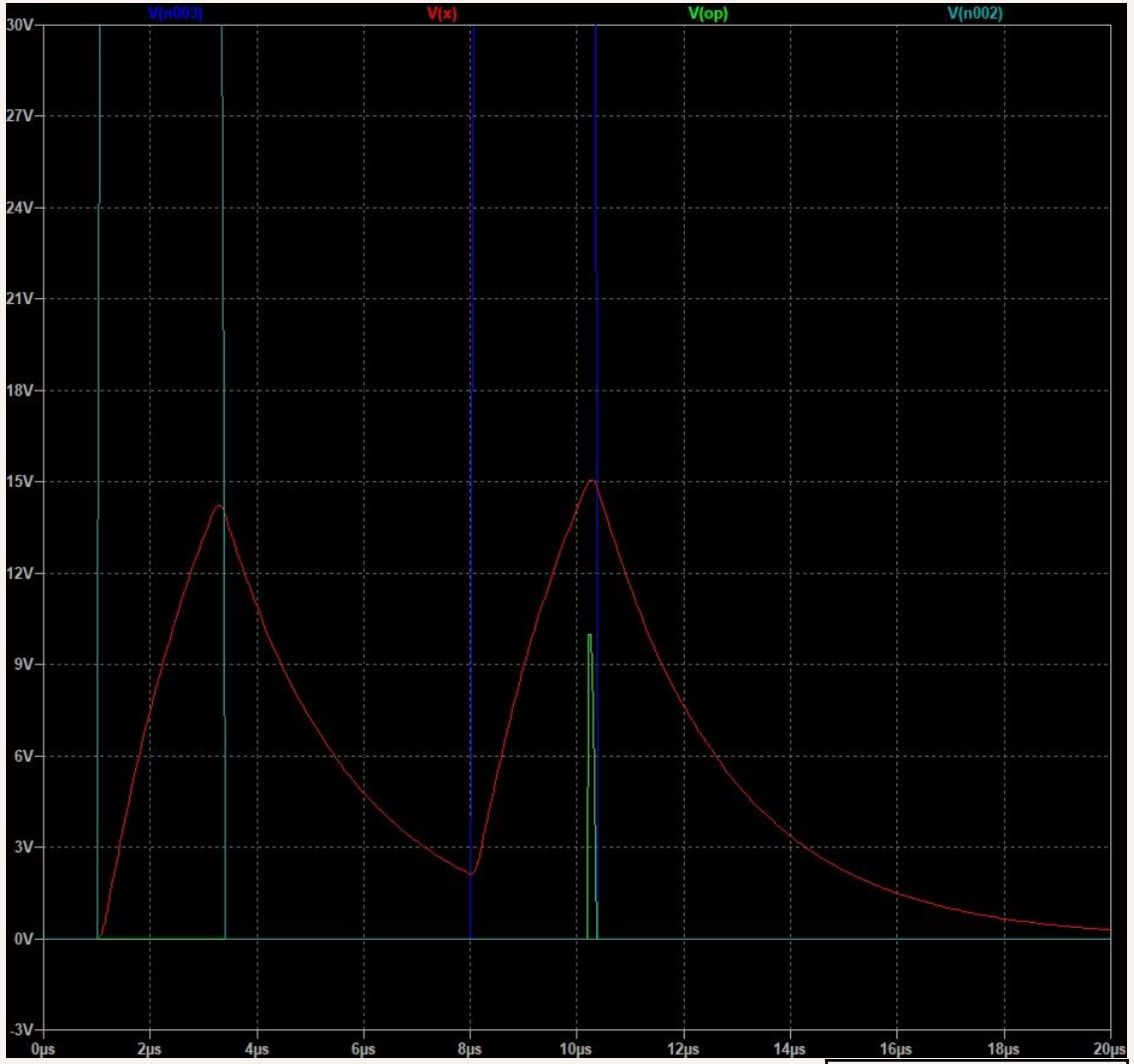

output
R_1
R_f
+
−
Vref
C
R_L
R_2
V_x

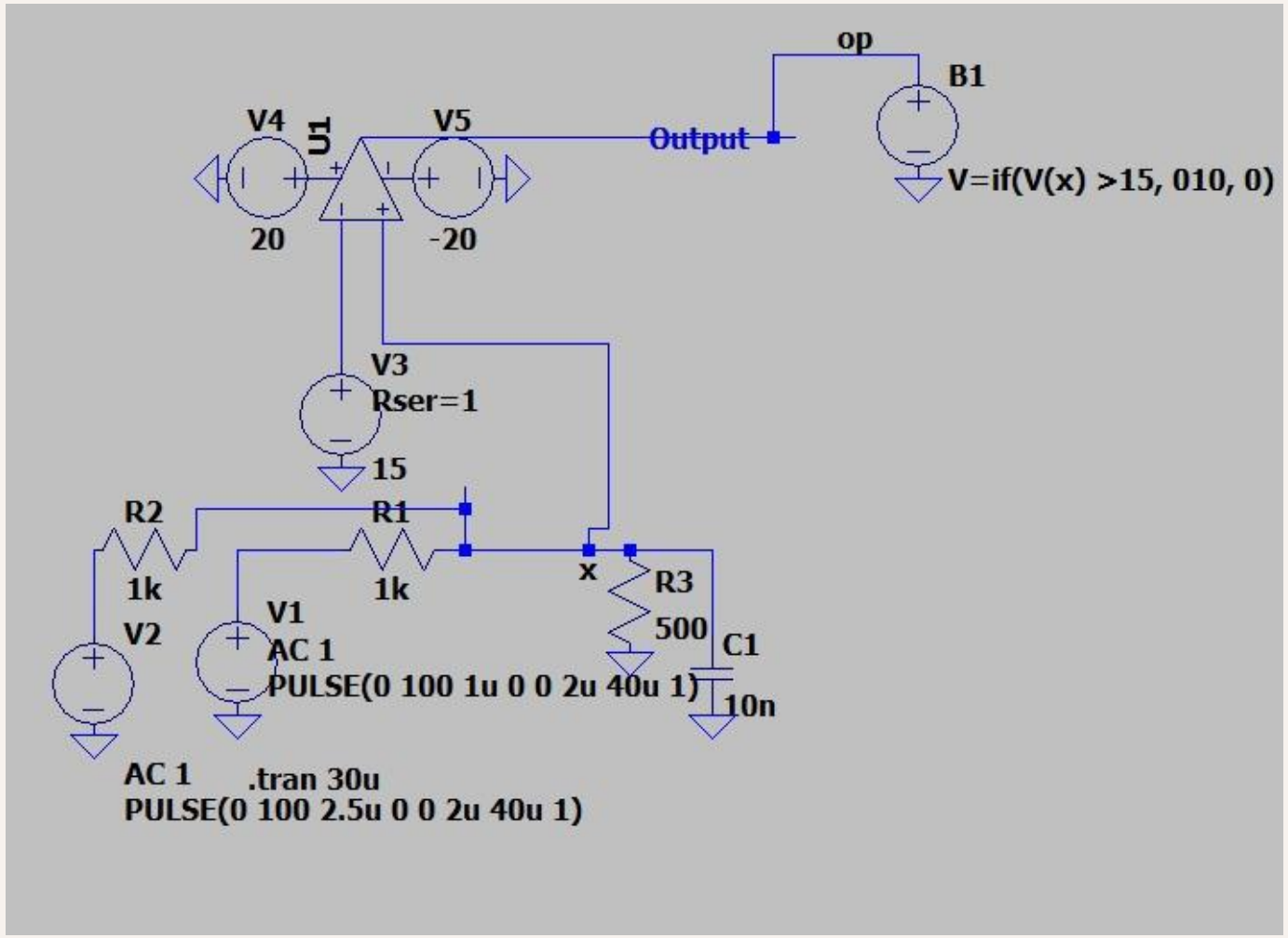
Plan to simulate coincidence detection in LTspice

1. Place two PULSE sources and feed them through R1 and R2 into the common node Vx.
2. Add RL + C at Vx.
3. Add buffer and comparator with Vref.
4. Set Pulse A delay = 0, Pulse B delay = TD (parameter).
5. Run .tran simulation and probe Vx + comparator output.
6. Use .step param TD to vary timing gap.
7. See at which TD the comparator fires the coincidence window.
8. Adjust RL, C if you want a bigger/smaller window.

least peak till signal can be fired

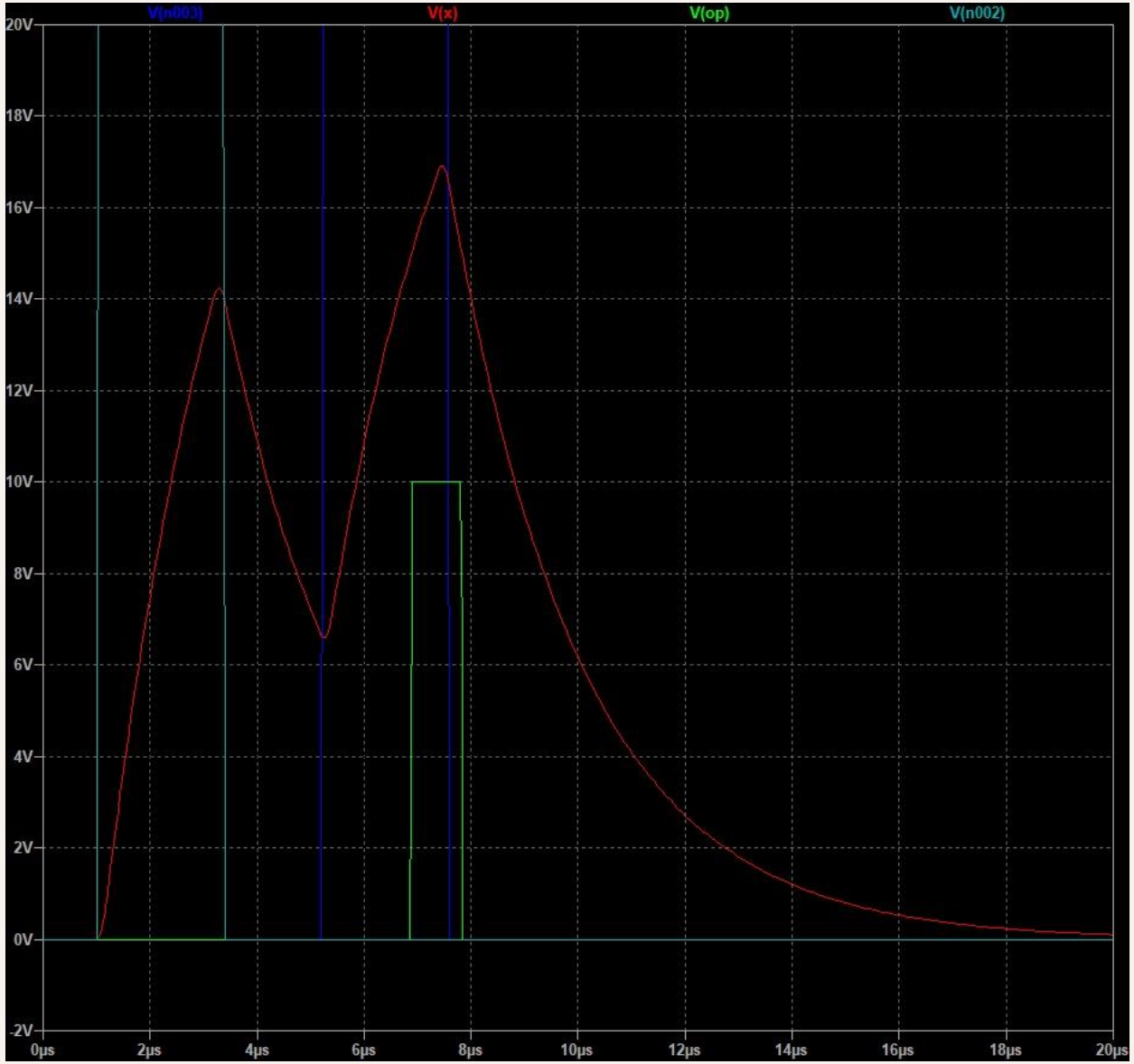
CIRCUIT



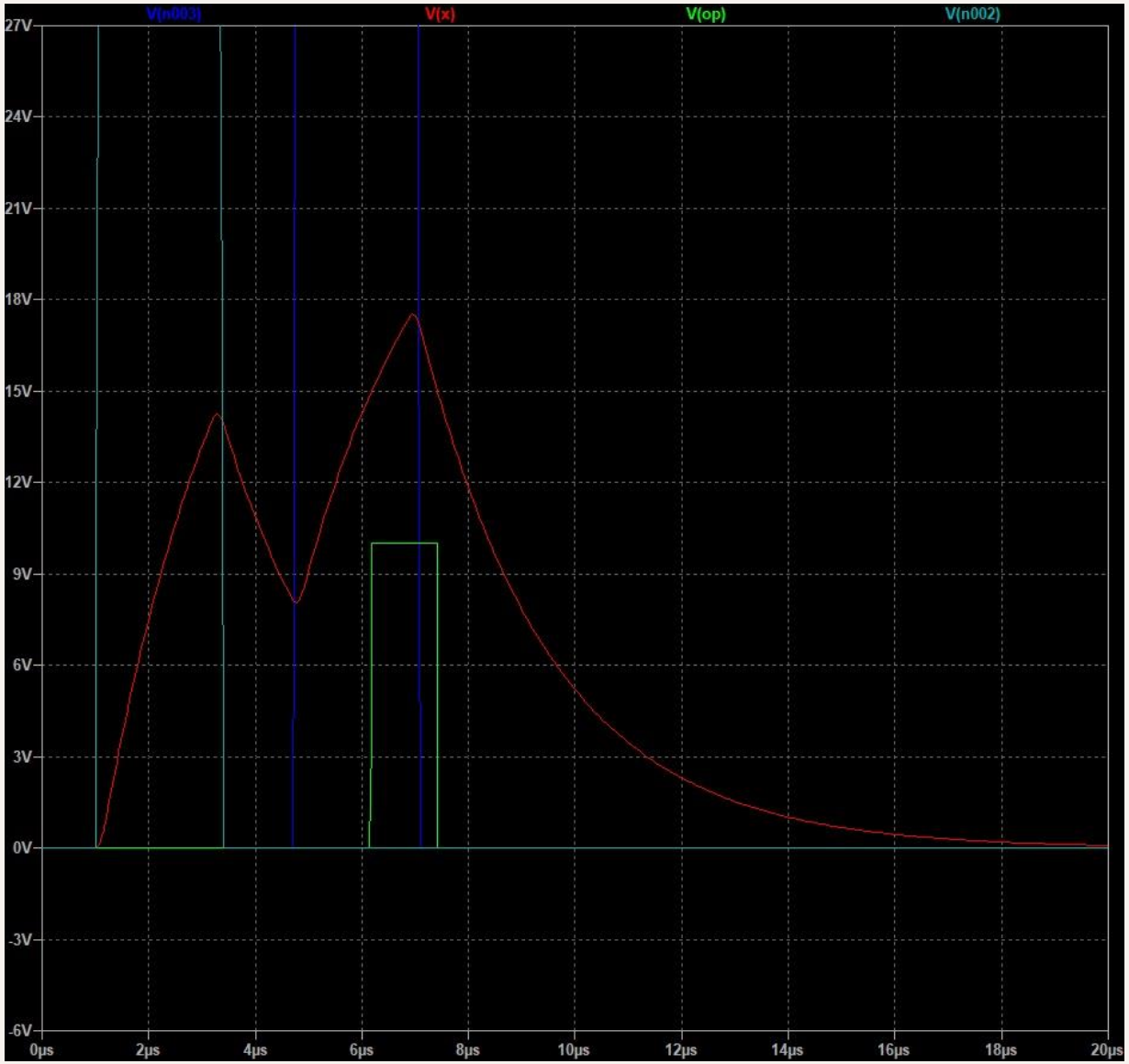


dt=T1+5

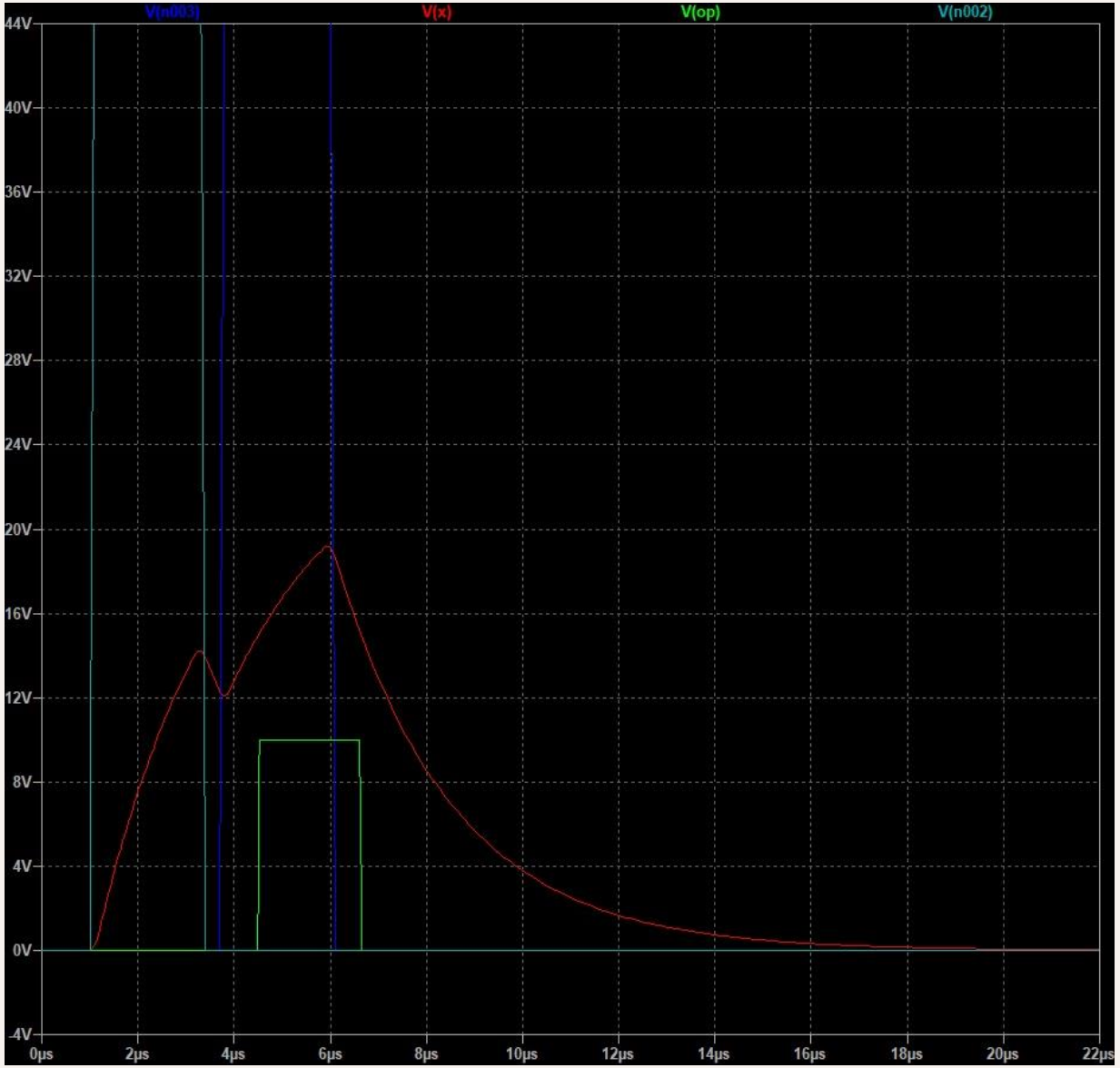
dt=T1+2



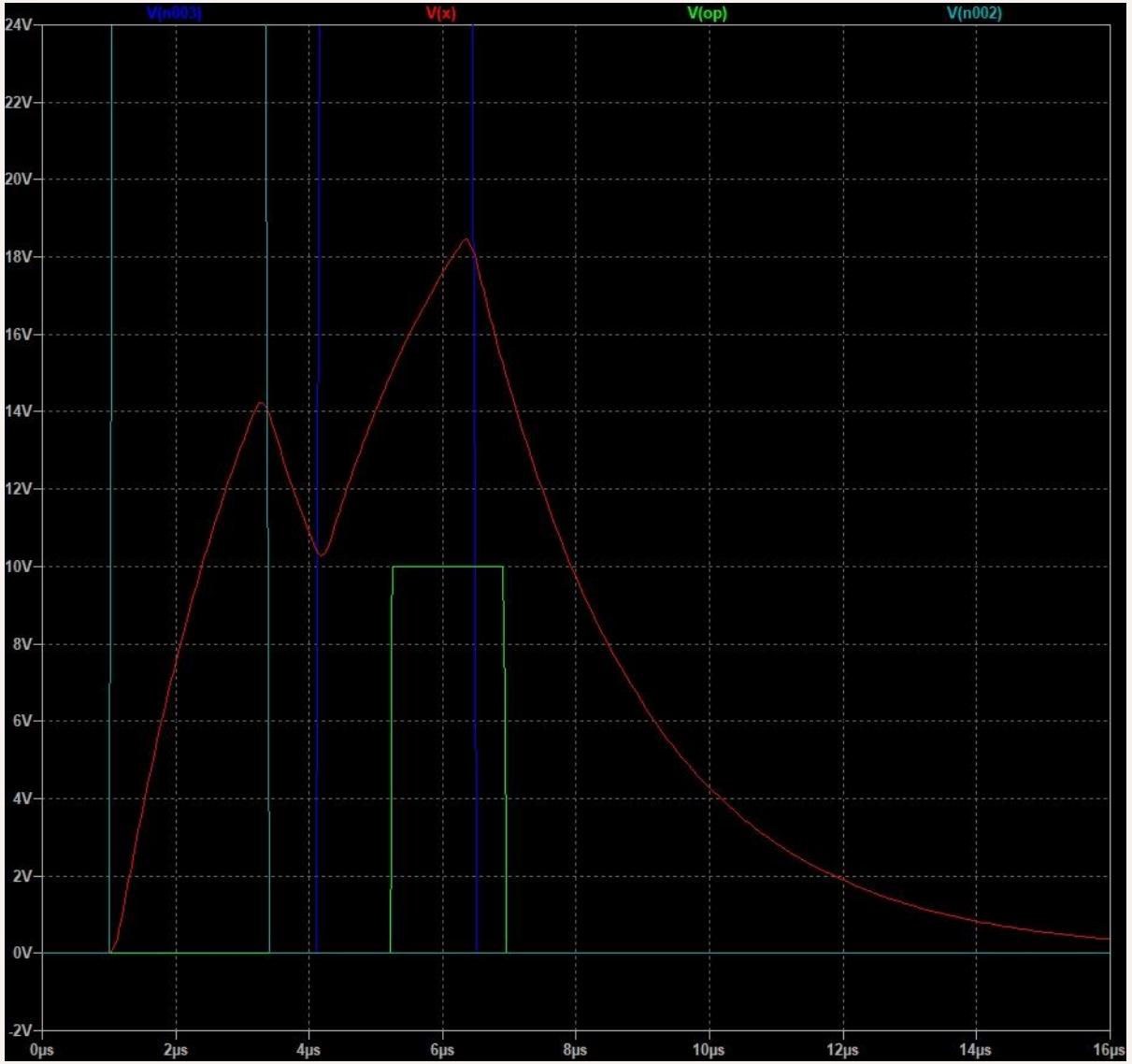
dt=T1+1.5



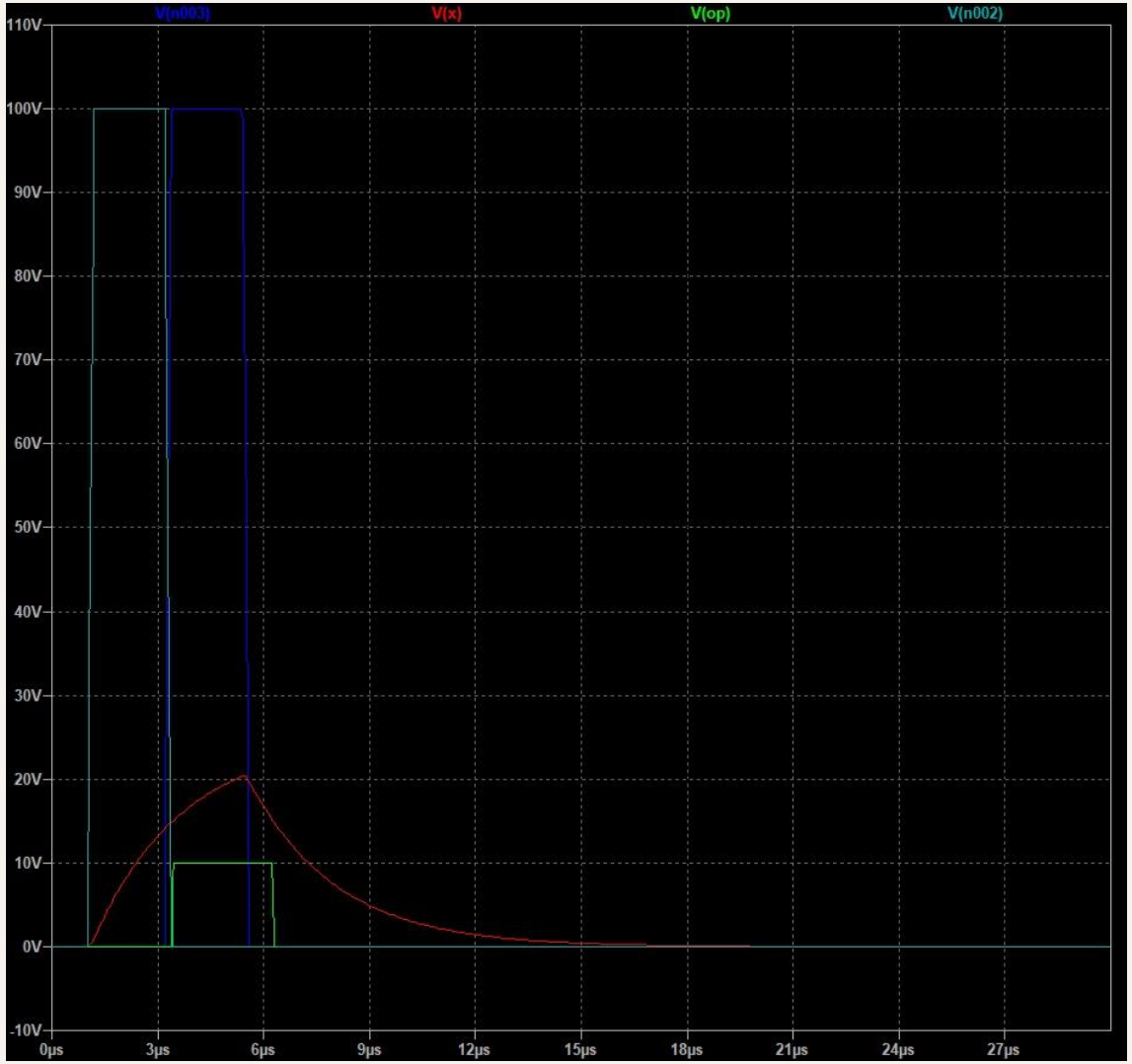
dt=T1+0.5 u



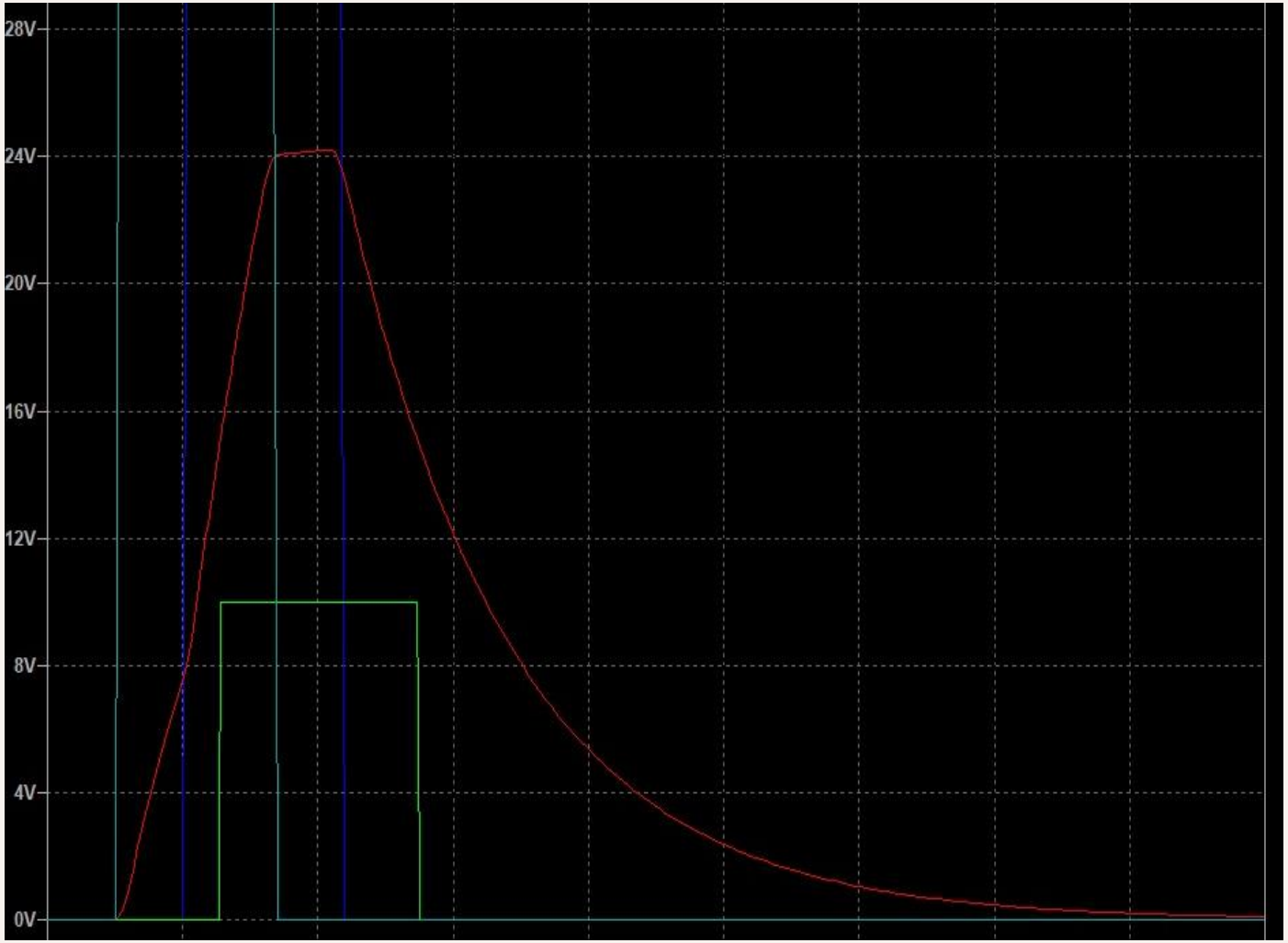
dt=T1+1



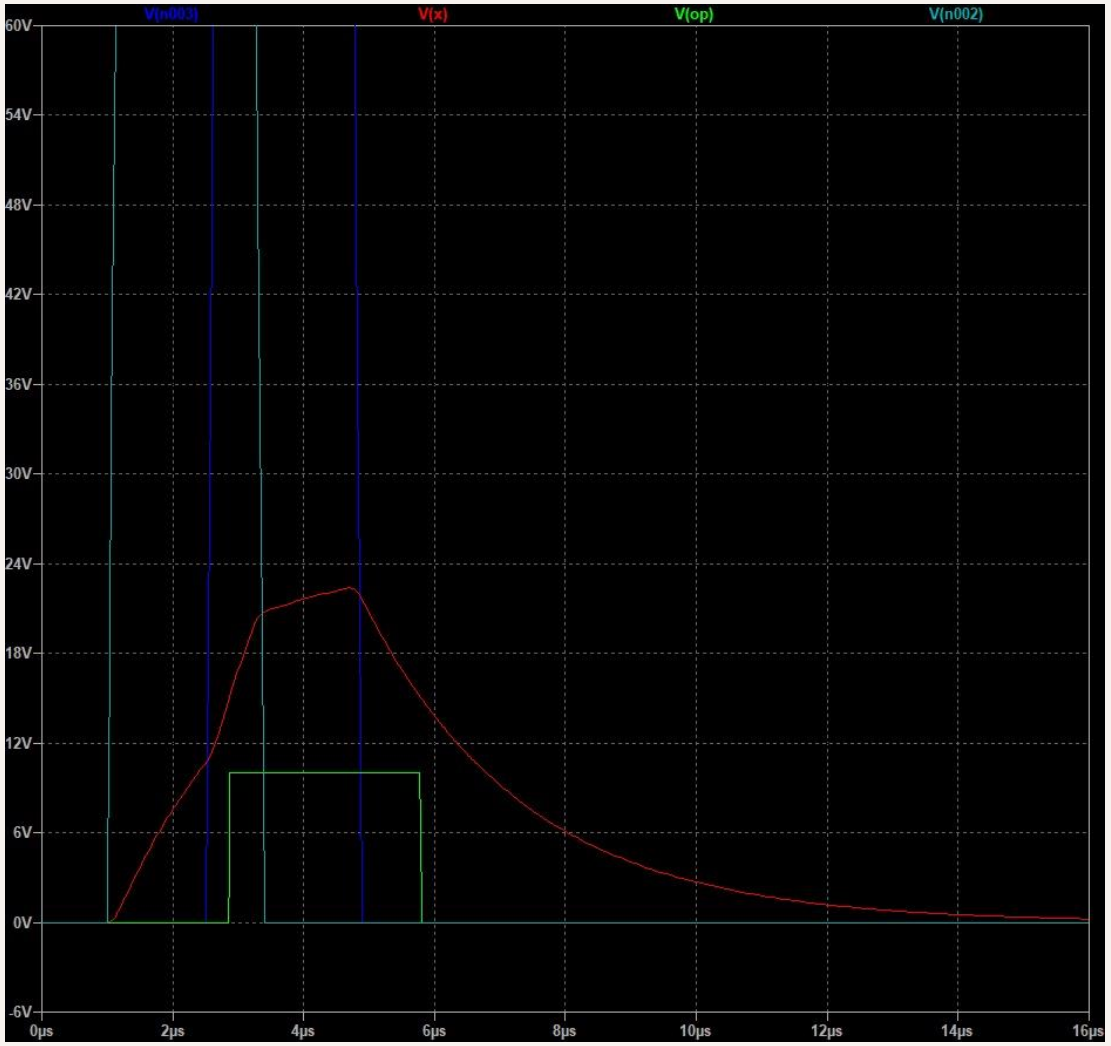
dt=T1


V(n003)
V(x)
V(op)
V(n002)
110V
100V
90V
80V
70V
60V
50V
40V
30V
20V
10V
0V
-10V
0µs
3µs
6µs
9µs
12µs
15µs
18µs
21µs
24µs
27µs

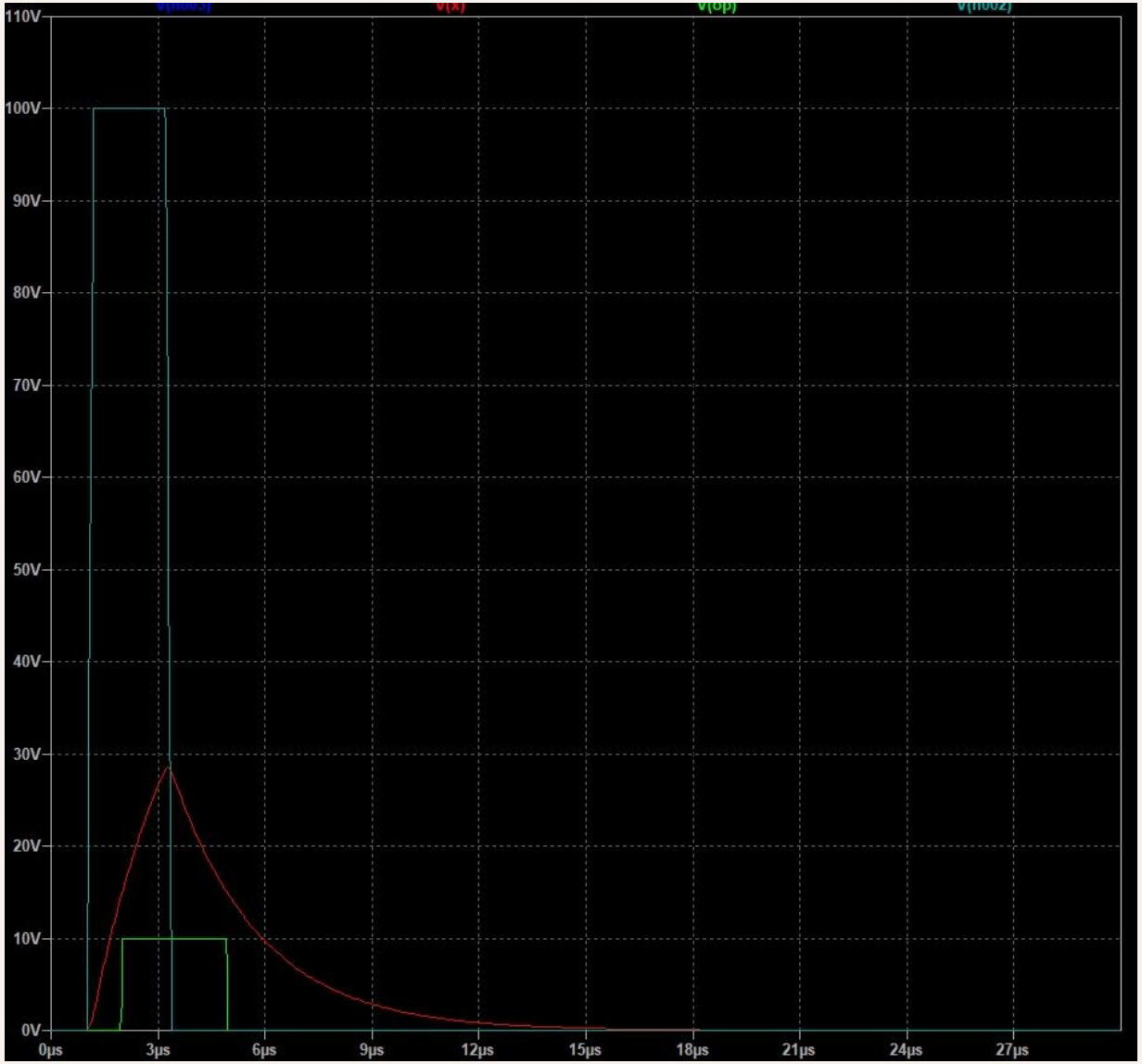
dt=1 us



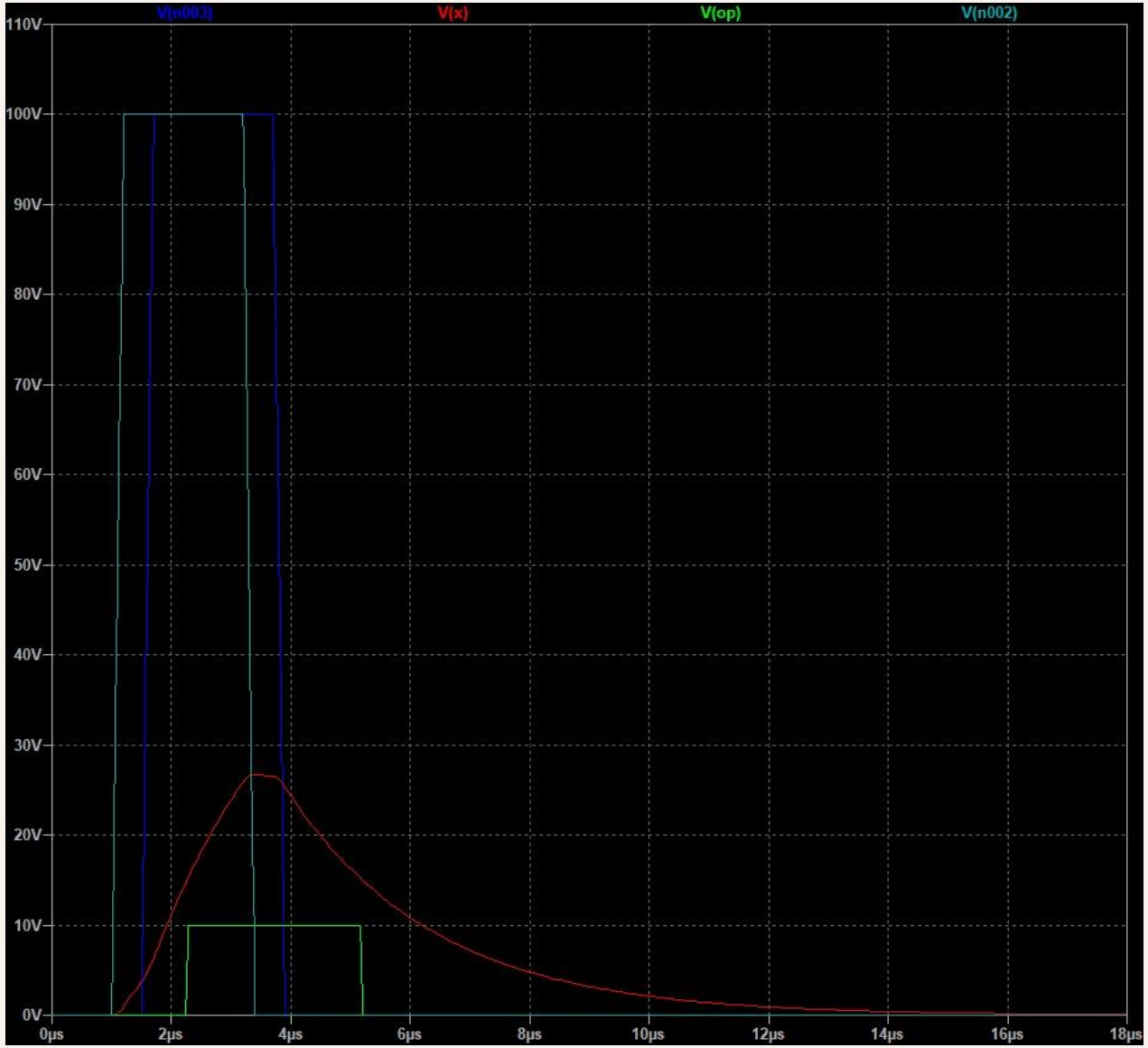
dt=1.5 us



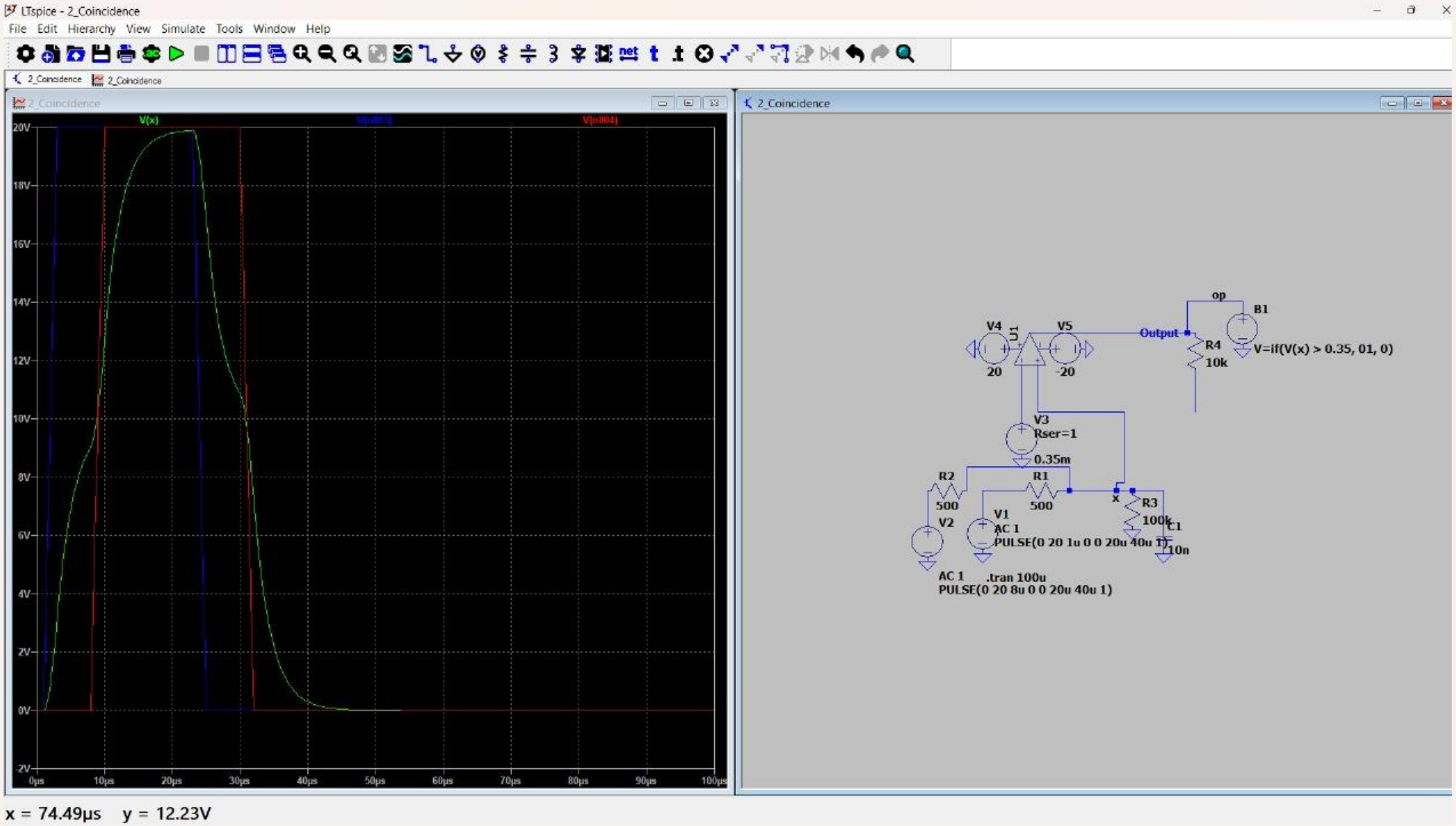
dt=0


V(x)
V(op)
110V
100V
90V
80V
70V
60V
50V
40V
30V
20V
10V
0V
0µs
3µs
6µs
9µs
12µs
15µs
18µs
21µs
24µs
27µs

dt=0.5 us


V(n003)
V(x)
V(op)
V(n002)
110V
100V
90V
80V
70V
60V
50V
40V
30V
20V
10V
0V
0µs
2µs
4µs
6µs
8µs
10µs
12µs
14µs
16µs
18µs

Problem


LTspice - 2_Coincidence
File Edit Hierarchy View Simulate Tools Window Help
2_Coincidence
V(x)
20V
18V
16V
14V
12V
10V
8V
6V
4V
2V
0V
-2V
0µs
10µs
20µs
30µs
40µs
50µs
60µs
70µs
80µs
90µs
100µs
x = 74.49µs y = 12.23V
op
B1
V4
V5
Output
R4
10k
V=if(V(x) > 0.35, 01, 0)
20
-20
V3
Rser=1
0.35m
R2
500
R1
500
x
R3
100k
C1
10n
V2
V1
AC 1
PULSE(0 20 1u 0 0 20u 40u 1)
AC 1 .tran 100u
PULSE(0 20 8u 0 0 20u 40u 1)